

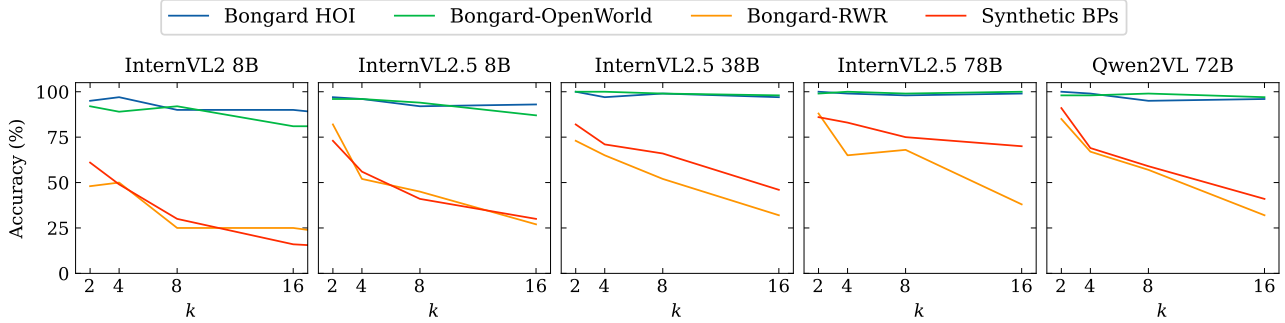


Figure 9: **Concept selection.** Multi-class classification accuracy of MLLMs across datasets and $k \in \{2, 4, 8, 16\}$ choices. Detailed results are presented in Table 3.

Table 3: **Concept selection.** Multi-class classification accuracy of MLLMs across datasets and $k \in \{2, 4, 8, 16\}$ choices. InternVL2 is denoted as IVL2, InternVL2.5 as IVL2.5, and Qwen2VL Instruct as Q2VL. High-level trends are illustrated in Fig. 9.

	BONGARD HOI				B-OPENWORLD				BONGARD-RWR				SYNTHETIC BPs			
	2	4	8	16	2	4	8	16	2	4	8	16	2	4	8	16
IVL2 8B	0.95	0.97	0.90	0.90	0.92	0.89	0.92	0.81	0.48	0.50	0.25	0.25	0.61	0.49	0.30	0.16
IVL2.5 8B	0.97	0.96	0.92	0.93	0.96	0.96	0.94	0.87	0.82	0.52	0.45	0.27	0.73	0.56	0.41	0.30
IVL2.5 38B	1.00	0.97	0.99	0.97	1.00	1.00	0.99	0.98	0.73	0.65	0.52	0.32	0.82	0.71	0.66	0.46
IVL2.5 78B	1.00	0.99	0.98	0.99	0.99	1.00	0.99	1.00	0.88	0.65	0.68	0.38	0.86	0.83	0.75	0.70
Q2VL 72B	1.00	0.99	0.95	0.96	0.98	0.98	0.99	0.97	0.85	0.67	0.57	0.32	0.91	0.69	0.59	0.41

D. The Impact of Model Scaling on Abstract Reasoning Abilities

Performance of MLLMs on downstream tasks is often correlated with the number of model parameters and the size of training datasets (Kaplan et al., 2020; Hoffmann et al., 2022). To investigate the relationship between model scaling and abstract reasoning performance, we conducted experiments with a diverse set of model sizes across proprietary and open-access MLLMs. To this end, we evaluated both smaller and larger variants of the selected models. Specifically, we considered GPT-4o mini and Gemini 1.5 Flash as smaller counterparts to GPT-4o and Gemini 1.5 Pro, resp. Also, we tested multiple configurations of InternVL2 and LLaVA-NeXT model families including InternVL2-8B, InternVL2-26B, InternVL2-40B, InternVL2-Llama3-76B, LLaVA-v1.6 Vicuna-13B, LLaVA-v1.6 34B, LLaVA-NeXT 72B, and LLaVA-NeXT 110B. We conducted experiments on all 4 datasets using two solution strategies, including Direct, which is an intuitive baseline, and Descriptive, the most effective strategy identified in the main experiments.

The results are presented in Fig. 10. In general, larger proprietary models outperformed their smaller counterparts in 10 out of 16 cases. However, smaller variants sometimes performed better than larger ones. For instance, on Bongard HOI with the Direct strategy, GPT-4o mini and Gemini 1.5 Flash surpassed their larger alternatives. This suggests that smaller models can achieve competitive performance in abstract reasoning.

For open-access models, performance consistently improved with model size. For example, the results of InternVL2 on Bongard HOI increased from 12 to 25 and from 2 to 29 for Direct and Descriptive strategies, resp. Similarly, on Bongard HOI, the performance of LLaVA-NeXT improved from 5 to 27 and from 4 to 27 for the two strategies. Analogous improvements were observed on Bongard-OpenWorld, highlighting the potential benefits of model scaling.

Despite these significant improvements in open-access models, proprietary models consistently outperformed them. In particular, GPT-4o mini achieved worse results than the best open-access model in a single case only, i.e., Bongard HOI using the Descriptive strategy (26 vs. 29). Although model scaling demonstrates its potential to enhance abstract reasoning, as shown by the open-access models, the relatively strong performance of GPT-4o mini shows that a large parameter count is not necessarily critical for excelling in abstract reasoning tasks. Consequently, these results suggest that simply scaling model size may be insufficient to achieve stronger abstract reasoning capabilities and future efforts should explicitly address this aspect, e.g., by incorporating AVR datasets into model training.

Table 4: **Evaluation results.** The number of correct answers to the first 100 synthetic BPs, 100 selected BPs from Bongard HOI and Bongard-OpenWorld, and all 60 BPs from Bongard-RWR. The best result for a given strategy is marked in bold, and the second best is underlined.

SYNTHETIC BPs	GPT-4 O	GPT-4 TURBO	GEMINI 1.5 PRO	CLAUDE 3.5 SONNET	INTERNVL2 8B	LLaVA-1.6 MISTRAL 7B	PHI 3.5V	PIXTRAL 12B
GROUND-TRUTH	<u>87</u>	48	78	84	30	70	22	92
INCORRECT LABEL	<u>79</u>	89	70	74	78	24	<u>79</u>	27
IMAGES TO SIDES	<u>72</u>	56	69	75	32	33	4	57
DIRECT	17	6	7	<u>13</u>	0	0	0	1
DESCRIPTIVE	17	15	21	<u>19</u>	0	1	2	4
DESCRIPTIVE-ITERATIVE	9	7	4	8	1	0	1	2
DESCRIPTIVE-DIRECT	22	13	20	22	1	0	1	5
CONTRASTIVE	10	8	17	<u>15</u>	0	0	0	1
CONTRASTIVE-ITERATIVE	20	9	<u>12</u>	11	1	0	0	4
CONTRASTIVE-DIRECT	<u>19</u>	9	20	18	1	0	1	1
BONGARD HOI	GPT-4 O	GPT-4 TURBO	GEMINI 1.5 PRO	CLAUDE 3.5 SONNET	INTERNVL2 8B	LLaVA-1.6 MISTRAL 7B	PHI 3.5V	PIXTRAL 12B
GROUND-TRUTH	73	<u>74</u>	60	32	41	70	49	100
INCORRECT LABEL	100	100	96	100	90	32	67	33
IMAGES TO SIDES	99	92	95	99	13	0	7	86
DIRECT	35	22	23	5	12	5	1	<u>28</u>
DESCRIPTIVE	42	45	40	<u>44</u>	2	4	4	27
DESCRIPTIVE-ITERATIVE	32	23	6	6	0	1	2	11
DESCRIPTIVE-DIRECT	45	<u>40</u>	30	29	7	3	7	25
CONTRASTIVE	18	5	<u>15</u>	13	2	1	2	7
CONTRASTIVE-ITERATIVE	22	20	15	<u>21</u>	2	1	3	14
CONTRASTIVE-DIRECT	<u>25</u>	12	27	15	7	0	1	7
BONGARD-OPENWORLD	GPT-4 O	GPT-4 TURBO	GEMINI 1.5 PRO	CLAUDE 3.5 SONNET	INTERNVL2 8B	LLaVA-1.6 MISTRAL 7B	PHI 3.5V	PIXTRAL 12B
GROUND-TRUTH	<u>80</u>	52	59	37	49	72	31	99
INCORRECT LABEL	100	100	98	99	86	44	83	29
IMAGES TO SIDES	94	86	96	96	19	2	9	87
DIRECT	40	21	13	10	11	12	7	<u>33</u>
DESCRIPTIVE	46	57	32	<u>53</u>	18	16	12	34
DESCRIPTIVE-ITERATIVE	31	<u>24</u>	6	13	4	2	2	11
DESCRIPTIVE-DIRECT	52	52	25	46	24	8	9	38
CONTRASTIVE	<u>19</u>	12	11	21	7	1	5	14
CONTRASTIVE-ITERATIVE	<u>25</u>	21	8	34	5	1	3	18
CONTRASTIVE-DIRECT	35	25	17	22	12	4	4	<u>27</u>
BONGARD-RWR	GPT-4 O	GPT-4 TURBO	GEMINI 1.5 PRO	CLAUDE 3.5 SONNET	INTERNVL2 8B	LLaVA-1.6 MISTRAL 7B	PHI 3.5V	PIXTRAL 12B
GROUND-TRUTH	22	5	26	10	13	38	21	58
INCORRECT LABEL	59	60	52	60	53	21	47	9
IMAGES TO SIDES	15	17	19	<u>24</u>	16	0	1	26
DIRECT	5	1	<u>3</u>	1	0	0	0	1
DESCRIPTIVE	<u>8</u>	5	7	13	0	0	0	1
DESCRIPTIVE-ITERATIVE	4	1	0	<u>3</u>	0	0	0	0
DESCRIPTIVE-DIRECT	5	7	5	<u>6</u>	0	0	0	2
CONTRASTIVE	2	0	1	2	0	0	0	0
CONTRASTIVE-ITERATIVE	5	3	3	<u>4</u>	0	0	0	1
CONTRASTIVE-DIRECT	<u>4</u>	1	5	<u>4</u>	0	0	0	0

E. Evaluation of MLLMs Answers

Preliminary experiments revealed that proprietary MLLMs are generally much more effective in solving Bongard Problems than open, publicly-available MLLMs. Therefore, all efforts devoted to optimizing the final scores, in particular tuning the evaluation prompt were performed using these 4 commercial MLLMs.

Open-ended characteristics of BPs stemming from a textual form of an answer, and the number of considered models (8),

Prompt 2: Initial prompt used in MLLM answer evaluation. We focused on its clarity and simplicity.

You are a logic module designed to provide accurate answers. In a Bongard Problem the
 → objective is to spot the difference between the contents of images located on the
 → two opposite sides of the problem. You are given correct labels of these sides and
 → must decide whether the answer provided by the user is correct and matches with
 → those labels. Answer with 'OK' or 'WRONG'.

LEFT SIDE LABEL:
 <left_label>

RIGHT SIDE LABEL:
 <right_label>

USER ANSWER:
 <model_answer>

generation strategies (7), datasets (4), and BP instances per dataset (60 in Bongard-RWR and 100 in the remaining cases) require the use of an automated NLP-based evaluation of the model’s answers. For this task we employed MLLMs with a specially designed prompt. The initial version of the evaluation prompt (see Prompt 2) was intentionally relatively simple – a model received an answer to be evaluated as well as the ground-truth labels, and was requested to output a binary *yes/no* answer. This prompt formulation turned out to be too simplistic. While the level of agreement between all models was relatively high (87% of responses were rated unanimously by all models), as illustrated in Fig. 11a, manual inspection of the selected answers revealed that the assessment was generally too optimistic and relatively many evaluations wrongly pointed to *correct* answers.

E.1. Evaluation Prompt Optimization

Due to the above evaluation disagreement, we made an attempt to optimize the prompt based on the GPT-4o solutions for the additional 20 BPs (#101 – #120) that were not used in the main experiments. First, following the few-shot prompting technique, we expanded the prompt to include two examples showing a possible logical difference between correct and incorrect answers. Furthermore, we added a sentence which requested a *strict* logical compliance with the provided labels. However, this refinement appeared to be too strong, as 2 (out of 4) models didn’t evaluate any of the solutions as correct.

To impose some flexibility, we changed the word *strictly* to *logically*, but this resulted in an increased rate of false-positive evaluations. Finally, we combined these two prompts, obtaining the outcome closest to the manual (our human) evaluation. The final version of our evaluation prompt is listed in Prompt 3. Additionally, we attempted to attach the image of the evaluated BP instance to each version of our prompt, but this actually confused the models rather than improving their results, so we ultimately abandoned this option and stuck with the fully text-based prompt.

Although the consistency of results regarded as the number of unanimous assessments stayed at the same level (87%) (see Fig. 11b), the number of answers rated as correct significantly decreased, which was in accordance with our random manual verification.

Despite lowering the results variation, there were still BPs for which the assessment varied. Therefore, we eventually decided to use **hard voting** to ensemble all models’ evaluations. We marked a solution as *correct* if at least 2 of the 4 models evaluated it as correct. This approach brought better results than the majority voting.

E.2. Manual Verification of the Evaluation Performance of MLLMs

In order to finally assess the efficacy of Prompt 3 we manually checked the models’ evaluation performance on the 100 BPs solved by GPT-4o using the *Descriptive* strategy. As shown in Table 4 in the main paper, all proprietary models achieved better scores on incorrect labels classification. For this reason, we decided to manually verify only the problems evaluated as correct by at least one MLLM, assuming that those incorrect are generally evaluated properly. The comparison between the evaluation performance of the initial and final prompts and our manual evaluation is presented in Table 5. *All models* denotes evaluations where a solution is marked as correct only if all models evaluate it as correct. Similarly, *any model*